



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL LOGISTICS

MAKE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

## SOVEREIGN UNMANNED AERIAL SYSTEMS PROGRAM

# INTEGRATED AUTONOMOUS SWARM & HYBRID LOGISTICS VTOL

A dual-pillar indigenous aerial capability engineered for contested Indian forward operations—delivering continuous multi-tier ISR swarm reconnaissance paired with runway-independent, heavy-cargo tactical logistics for high-altitude border resupply.

## PILLAR 01 TACTICAL SWARM &amp; ISR



## SWARM & OVERWATCH ISTAR

[ SECT 01 ]

MULTI-LAYER DISTRIBUTED SURVEILLANCE &amp; MESH C2

Coordinated heterogeneous swarm linking disposable micro Nano-scouts (350–450g) and long-endurance Overwatch UAV relays (~20kg) via self-healing peer-to-peer data-links.

- Decentralized autonomous collision avoidance and patrol logic
- Real-time EO/IR thermal target tracking streamed to tactical GCS
- Attrition-tolerant multi-node coordination in contested airspace

SWARM NODE SIZE

6 to 24+ Units

OPERATING CEILING

8,000 ft AMSL

OVERWATCH MTOW

~20 kg Class

C2 DATA-LINK

Secure Mesh RF

## PILLAR 02 DEFENCE LOGISTICS VTOL



## HYBRID LOGISTICS VTOL

[ SECT 02 ]

RUNWAY-FREE TACTICAL RESUPPLY &amp; MEDICAL TRANSIT

All-terrain hybrid VTOL + fixed-wing platform engineered for forward post resupply, trauma medical delivery, and ammunition transport in rugged Himalayan sectors without prepared runways.

- Vertical takeoff and transition to efficient fixed-wing cruise
- Quick-change modular cargo pod for field-swappable payloads
- Emergency parachute failsafe and precision winch drop capability

PAYLOAD CAPACITY

5-10 to 200 kg

CRUISE FLIGHT

Up to 180 Min

MISSION RANGE

100+ km Transit

NAVIGATION

NAVIC + INS Ready

## DUAL-SECTOR OPERATIONAL DOCTRINE

UNIFIED INDIAN DEFENCE ARCHITECTURE

- Common Tactical GCS: Single operator console for both Swarm & Cargo assets
- Contested Spectrum Survivability: NAVIC satellite sync + frequency-hopping RF
- High-Altitude Deployment: Sub-zero temperature and mountain turbulence rated
- Make in India Manufacturing: Sovereign IPR with localized assembly & maintenance

### 100% Runway-Free

VTOL OPERATIONAL ENVELOPE

Autonomous vertical takeoff and landing from confined outpost pads, warships, or mountain clearings.

### 50-150 km

OPERATIONAL NETWORK RADIUS

BVLOS encrypted telemetry with multi-hop swarm mesh routing for contested spectrum environments.

### 5 to 200 kg

MODULAR PAYLOAD ARCHITECTURE

Quick-change bays for medical trauma boxes, tactical resupply pods, or stabilized EO/IR surveillance gimbals.

### Make in India

SOVEREIGN IPR &amp; ASSEMBLY

Complete indigenous production roadmap aligned with IDEX and Ministry of Defence procurement guidelines.